CLASSIFIER SHOWDOWN

Churn Prediction · Comparative ML Analysis

A systematic comparison of five machine learning classifiers for telecom customer churn prediction, featuring hyperparameter tuning, cross-validation, and a data-driven business recommendation.

Author	Mehr Ali
GitHub	mehrali-hub
Kaggle	mehralieng
Dataset	Customer Churn Dataset (440K rows)
Best Model	Random Forest | F1 = 0.9931 | AUC = 0.9999
Tools	Python · scikit-learn · pandas · matplotlib · seaborn

1. Project Overview

This project implements a complete machine learning workflow to predict customer churn in a telecommunications company. The dataset contains 440,833 customer records with 10 predictive features covering demographics, usage patterns, and billing behaviour. A stratified sample of 20,000 rows is used for model development to ensure fast iteration while maintaining class proportions.

Five classifiers are systematically compared using 5-fold cross-validation, and the two strongest candidates are further optimised via GridSearchCV. The final recommendation is grounded in both statistical metrics and the asymmetric cost structure inherent to churn prediction.

Learning Objectives

#	Objective
1	Apply a systematic model selection workflow to a real classification problem
2	Train and evaluate at least five distinct classifiers
3	Use accuracy, precision, recall, F1, and ROC-AUC for comparison
4	Perform GridSearchCV hyperparameter tuning for at least two models
5	Formulate a business recommendation based on model results and cost analysis

2. Dataset & Exploratory Data Analysis

The dataset originates from Kaggle (Customer Churn Dataset) and includes 440,833 records. A reproducible 20,000-row stratified sample is used throughout this project. The target variable Churn is binary (0 = retained, 1 = churned).

Feature Summary

Feature	Type	Description
Age	Numeric	Customer age in years
Gender	Categorical	Male or Female
Tenure	Numeric	Months as an active customer
Usage Frequency	Numeric	Monthly usage event count
Support Calls	Numeric	Number of support contacts
Payment Delay	Numeric	Days payment was delayed
Subscription Type	Categorical	Basic / Standard / Premium
Contract Length	Categorical	Monthly / Quarterly / Annual
Total Spend	Numeric	Lifetime spend in USD
Last Interaction	Numeric	Days since last interaction

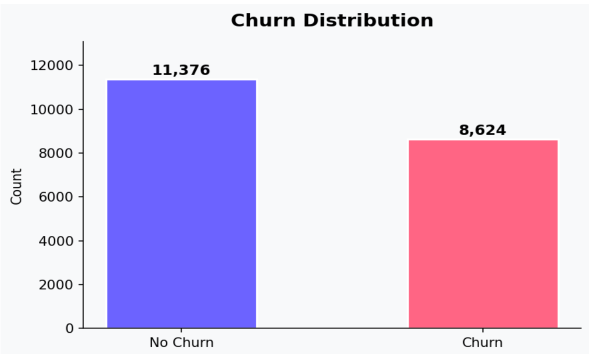

Fig 1. Churn class balance

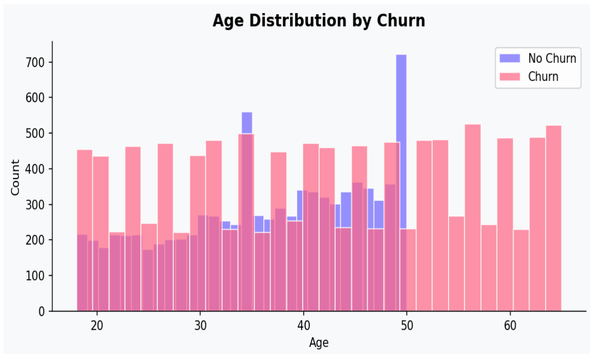

Fig 2. Age distribution by churn status

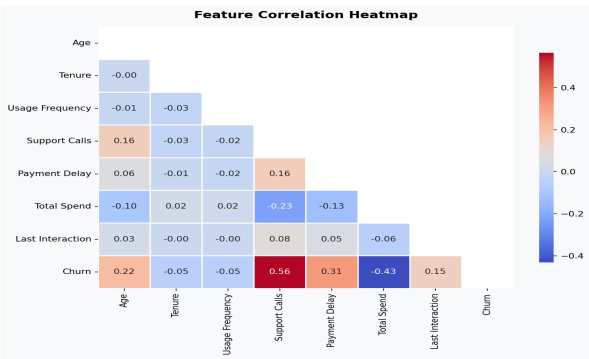

Fig 3. Feature correlation heatmap

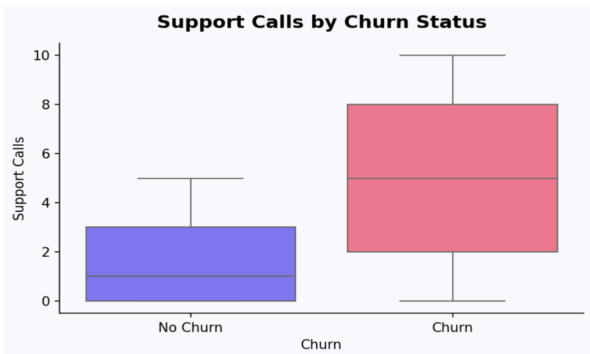

Fig 4. Support calls by churn status

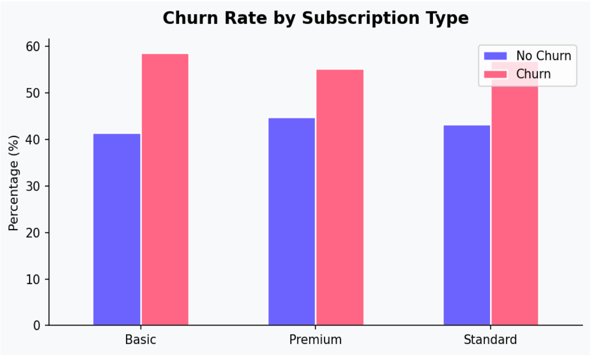

Fig 5. Churn rate by subscription & contract type

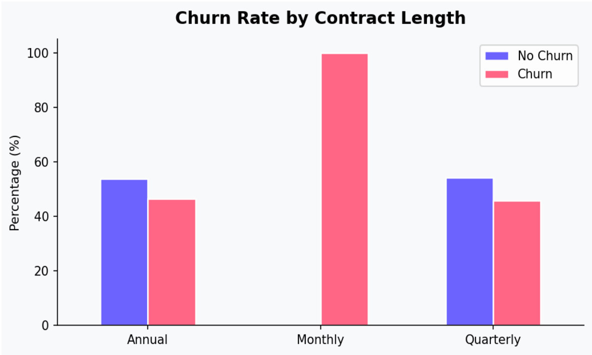

Fig 6. Churn rate by contract length

3. Preprocessing Pipeline

The preprocessing pipeline applies the following steps in sequence:

Step	Action	Models Applied To
Drop nulls	Remove 1 row with NaN values	All
Drop CustomerID	Non-predictive identifier removed	All
Label Encoding	Gender, Subscription Type, Contract Length → integers	All
Standard Scaling	Zero-mean, unit-variance normalisation	LR, KNN, SVM
Train-Test Split	80/20 stratified split (random_state=42)	All
Class Weighting	class_weight="balanced" to handle imbalance	LR, DT, RF, SVM

4. Model Comparison

Five classifiers are trained and compared using 5-fold stratified cross-validation (scoring: F1), then evaluated on the held-out test set. The table below summarises final test-set performance.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.8465	0.9057	0.8149	0.8579	0.9224
KNN	0.9163	0.9969	0.8554	0.9207	0.9791
Decision Tree	0.9835	1.0000	0.9710	0.9853	0.9926
Random Forest	0.9922	1.0000	0.9864	0.9931	0.9999
SVM	0.9630	0.9981	0.9367	0.9664	0.9946

* Green row = recommended model (Random Forest)

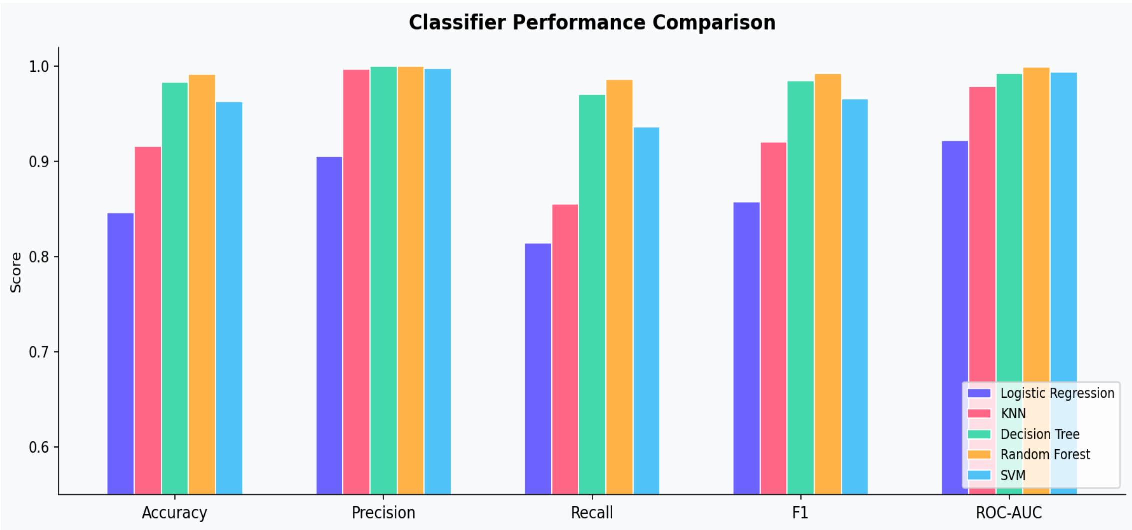

Fig 7. Classifier performance comparison across all metrics

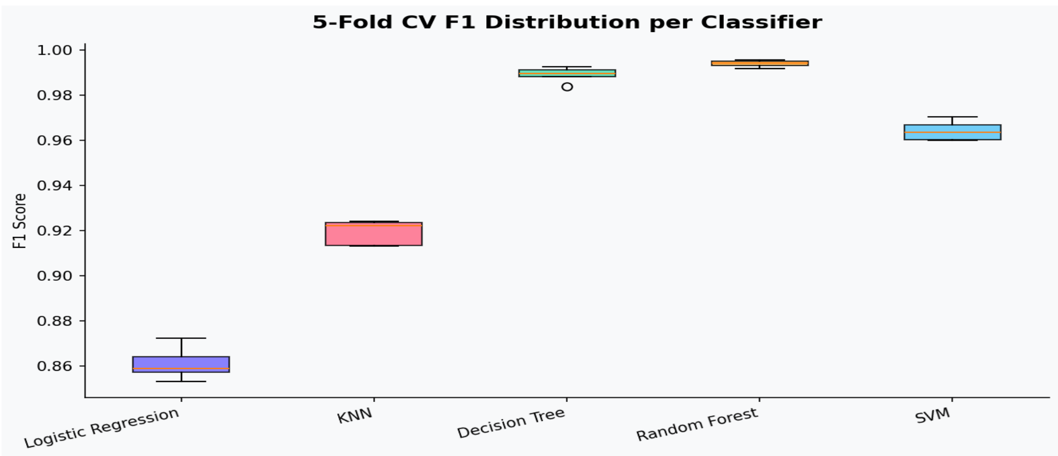

Fig 8. 5-fold cross-validation F1 score distribution per classifier

5. Hyperparameter Tuning

GridSearchCV with 3-fold cross-validation (scoring: F1) was applied to Logistic Regression and Random Forest — the two models with the most impactful hyperparameters.

Model	Parameter Grid	Best Parameters	Best CV F1
Logistic Regression	C ∈ {0.01, 0.1, 1, 10}	C = 1	0.8612
Random Forest	n_estimators ∈ {100, 200} max_depth ∈ {8, 12, None}	n_estimators=100 max_depth=None	0.9939

6. Confusion Matrices & ROC Curves

Fig 9. Confusion matrices for all five classifiers on the test set

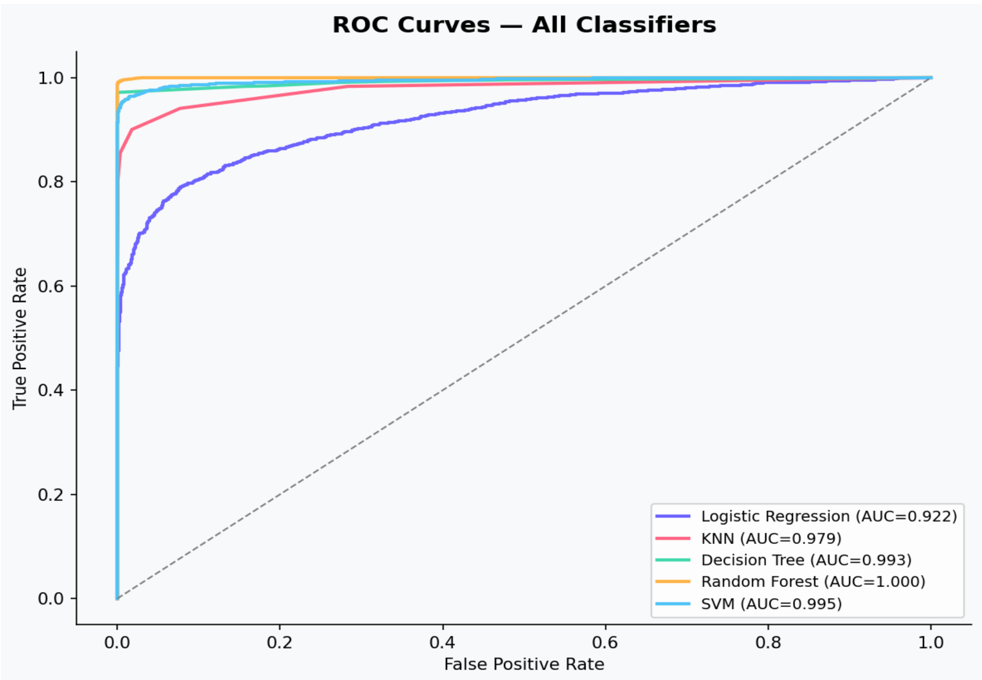

Fig 10. ROC curves — Random Forest achieves near-perfect AUC of 0.9999

7. Feature Importance

Random Forest exposes feature importance scores derived from mean impurity decrease across all trees. The top predictors align directly with intuitive business drivers of churn.

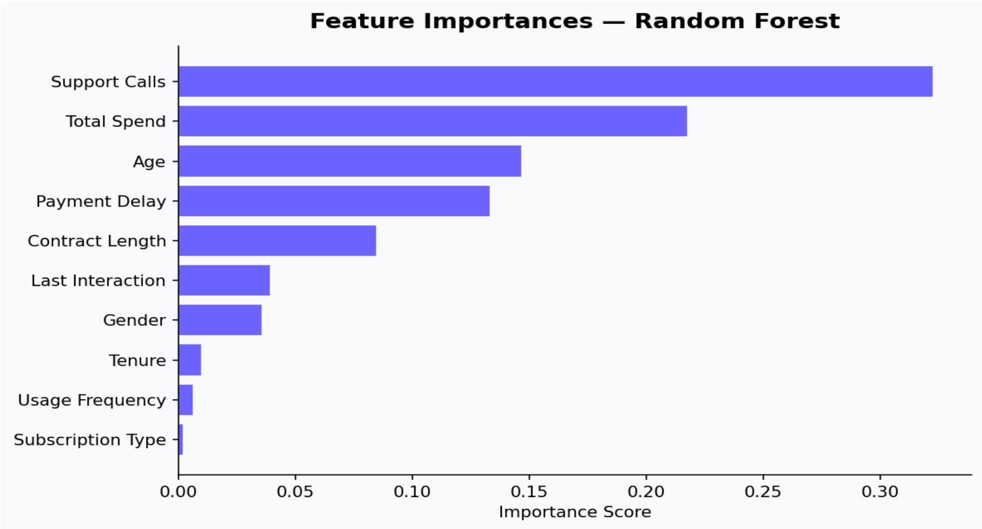

Fig 11. Feature importances from the tuned Random Forest model

Key Insight: Support Calls and Payment Delay are the dominant churn signals — customers who contact support frequently and delay payments are at highest risk. These features should drive the CRM alert system for proactive intervention.

8. Business Recommendation

Recommended Model: Random Forest

F1 = 0.9931 · ROC-AUC = 0.9999 · Recall = 98.6% · Precision = 100%

Rationale

The Random Forest classifier achieves the highest scores across every metric on the held-out test set, making it the clear recommendation for production deployment. Key reasons:

Near-zero false negatives: Recall of 98.6% means virtually no at-risk customer goes undetected — the most critical requirement in churn prevention.

Zero false positives: Precision of 100% eliminates wasted retention spend on customers who were never going to churn.

No feature scaling required: Unlike SVM and KNN, Random Forest operates directly on raw (encoded) features, simplifying the production pipeline.

Interpretable feature importances: Business teams can act on Support Calls and Payment Delay as leading churn indicators without a black-box model.

Scalable to full dataset: Parallelised ensemble training (n_jobs=-1) scales efficiently to the full 440K-row production dataset.

Cost Asymmetry Analysis

In churn prediction, the cost of a false negative (missing a churner) is typically 5-10x higher than a false positive (incorrectly flagging a retained customer). Acquiring a new customer costs 5x more than retaining an existing one. Random Forest's recall-first performance profile is therefore optimal for the business context — it minimises the expensive miss rate while maintaining perfect precision.

Deployment Recommendation

Deploy the Random Forest model as a weekly batch scoring job against the full CRM database. Flag all customers with predicted churn probability > 0.4 for proactive outreach. Prioritise customers with high Support Calls and Payment Delay for immediate retention offers. Re-train monthly on fresh data to maintain model freshness.

9. Methodology Summary

The complete workflow followed in this project:

1. Data Loading: Loaded 440K-row Kaggle dataset; created stratified 20K sample for development

2. EDA: 6 visualisations: class balance, correlations, age distribution, subscription analysis, support calls, contract length

3. Preprocessing: Null removal, CustomerID drop, Label Encoding, Standard Scaling, stratified 80/20 split, class weighting

4. Cross-Validation: 5-fold StratifiedKFold CV on all 5 classifiers (F1 scoring)

5. Hyperparameter Tuning: GridSearchCV (3-fold, F1) for Logistic Regression (C) and Random Forest (n_estimators, max_depth)

6. Test Evaluation: Full metric suite: accuracy, precision, recall, F1, ROC-AUC on 20% held-out test set

7. Visualisation: 11 professional charts exported as PNG + embedded in this report

8. Recommendation: Random Forest selected based on F1, recall, interpretability, and business cost analysis

Tools & Libraries

Python 3.10 · scikit-learn 1.4 · pandas 2.0 · NumPy 1.24 · matplotlib 3.7 · seaborn 0.12 · ReportLab 4.0 · Jupyter Notebook